

PID-Based Formation Cooperative Control for Quadrotor UAVs

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Abstract. With the development of UAV technology, quadrotor UAV formation control has become an important research direction for modern UAV applications. In this paper, the control system design and formation cooperative control technology of quadrotor UAV are studied. Using the MATLAB simulation platform, a control system based on PID controller is designed to realize the stable flight and precise control of the UAV. Meanwhile, a leader-follower formation cooperative control strategy is proposed to realize coordinated movement and formation keeping among multiple UAVs through reasonable control algorithms.

Keywords: Quadcopter UAV, Formation control, PID controller, Cooperative control.

1 Introduction

With the continuous progress of science and technology, UAV technology has developed rapidly and plays an important role in many fields such as military reconnaissance, environmental monitoring, and agricultural spraying. Quadrotor UAVs have occupied an important position in the field of UAVs due to their simple structure, flexible handling, and good stability¹. However, a single quadrotor UAV often has limitations when performing complex tasks, so it is of great theoretical and practical significance to study the multi-UAV formation cooperative control technology to improve the comprehensive application capability of UAS². In terms of control system design, this paper elaborates the design process of PID controller, including the parameter selection and adjustment methods of proportional, integral and differential³. In terms of formation cooperative control technology, this paper proposes a pilot-follower control strategy and designs a control algorithm based on distance and angle, which realizes the relative position maintenance and formation adjustment between UAVs in formation⁴. The research results in this paper not only provide theoretical basis and technical support for the control system design and formation cooperative control of quadrotor UAVs, but also are of great significance for promoting the application of UAV technology in a wider range of fields.

2 Quadcopter UAV modeling

A quadcopter drone, or quadcopter, is a vertical take-off and landing vehicle equipped with four propellers. The four propellers are symmetrically distributed in the four directions of the fuselage, with the motors mounted at the end of the bracket and the flight control computer and other equipment in the center. By adjusting the speed of the motors, the lift force is changed, thus controlling the attitude and position of the vehicle. A quadrotor is a typical underdriven system with six degrees of freedom but only four control inputs, making the control design complex. There is strong coupling between its control channels, e.g., pitch motion affects roll and yaw. Its nonlinear dynamics model and environmental factors increase the control difficulty, so the control system needs to be highly robust. In addition, how to guarantee the flight safety when the actuator fails is also an important research topic. In recent years, researchers have proposed various fault-tolerant control methods, such as integral sliding mode control, to enhance the safety and reliability of UAVs⁵.

In order to better describe the attitude of the UAV, we need to define two coordinate systems: one is the ground coordinate system and the other is the airframe coordinate system. First, let's establish the ground coordinate system. The ground coordinate system is a coordinate system fixed on the surface of the earth. We can choose any point O on the ground as the origin, and the X-axis points to any direction on the ground, the Z-axis points to the sky along the vertical direction, and the Y-axis is in the horizontal plane, perpendicular to the X-axis, and its direction is determined by the right-hand rule. Parameters such as the UAV's position, velocity and angular velocity are measured relative to this coordinate system. Therefore, we denote this ground coordinate system as {E-Oxyz} system as shown in Fig. 1(a).

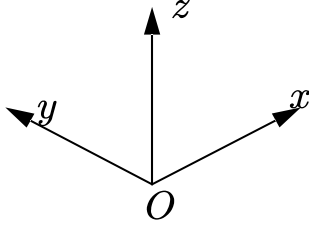


Fig. 1(a) {E-Oxyz} system

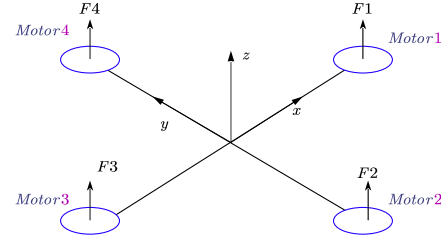


Fig. 1(b) B-Oxyz system

The airframe coordinate system is obviously built on the vehicle. Its origin O is located at the center of mass of the vehicle, and for a quadrotor UAV, the origin is actually the center of the quadrotor. The X-axis is located in the plane of symmetry of the aircraft and is parallel to the design axis of the vehicle, pointing forward of the nose; the Y-axis is perpendicular to the plane of symmetry of the fuselage, pointing to the right side of the fuselage; and the Z-axis passes through the point O and is perpendicular to the XOY plane, pointing to the upper part of the vehicle. This coordinate system is called the {B-Oxyz} system shown in Fig. 1(b).

2.1 Modeling of UAV dynamics

Modeling of UAV dynamics is primarily based on Newton's second law, which states that

$$\vec{F} = m \frac{d\vec{V}}{dt}, \vec{M} = \frac{d\vec{H}}{dt}, \quad (1)$$

where $\vec{F}$ is the sum of all external forces acting on the quadrotor; $\vec{V}$ is the quadrotor center of mass velocity; m is the quadrotor mass; and $\vec{H}$ is the momentum moment of the quadrotor relative to the ground coordinate system. Considering first the external forces acting on the quadrotor, then the following equation can be obtained: $m\mathbf{a} = F_D + F_G + f + D_i$, where $m\mathbf{a}$ denotes the acceleration of

the UAV position information, and F_D denotes the sum of the lift generated by the four rotors of the quadrotor UAV, the above equation is expressed in the ground coordinate system, so the sum of the lift generated by the four rotors of the quadrotor needs to be expressed in the ground coordinate system, defining that the sum of the lift generated by the four rotors of the quadrotor UAV is expressed as, ${}^B F_D$ under the fuselage coordinate system, then the magnitude is

${}^B F_D = \sum_{j=1}^4 \rho \omega_j^2$, where ρ is the lift coefficient and ω_j is the angular velocity of the motor rotation. The direction of this force is in the same direction as the z-axis of the fuselage coordinate

system, so it can be written as ${}^B F_D = \begin{bmatrix} 0 & 0 & \sum_{j=1}^4 \rho \omega_j^2 \end{bmatrix}^T$. Convert it to a terrestrial coordinate

system $F_D = {}^E F_D = {}^E R {}^B F_D$, where F_G denotes the gravitational force on the quadrotor, which is oriented vertically downward in the ground coordinate system and has a magnitude of mg ; f denotes the air resistance on the quadrotor, which has a magnitude of $f = kv$, k is the coefficient of air resistance, and v is the velocity of the vehicle; and D_i denotes the unmedicable internal part of the quadrotor UAV and the external perturbation, which is given by the expression:

$$D_i = \begin{bmatrix} d_1 & d_2 & d_3 \end{bmatrix}^T \quad (2)$$

It gives the following equation

$$m \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = {}^E R \begin{bmatrix} 0 \\ 0 \\ \sum_{j=1}^4 \rho \omega_j^2 \end{bmatrix} - m \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} - \begin{bmatrix} k_1 \dot{x} \\ k_2 \dot{y} \\ k_3 \dot{z} \end{bmatrix} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \quad (3)$$

Where k_1, k_2, k_3 are the drag coefficients x, y, z , which represent the velocities in the $\dot{x}, \dot{y}, \dot{z}$

directions, respectively, and substituting the value of ${}^E_B R$ into the above equation yields

$$\begin{cases} \ddot{x} = (\sum_{j=1}^4 \rho \omega_j^2 / m) (s\psi s\varphi + c\psi c\varphi s\theta) - (k_1/m)\dot{x} + d_1 \\ \ddot{y} = (\sum_{j=1}^4 \rho \omega_j^2 / m) (s\psi s\theta c\varphi + c\psi s\varphi) - (k_2/m)\dot{y} + d_2 \\ \ddot{z} = (\sum_{j=1}^4 \rho \omega_j^2 / m) c\theta c\varphi - g - (k_3/m)\dot{z} + d_3 \end{cases} \quad (4)$$

The above equation is the UAV position model, where $s\theta = \sin(\theta)$, $c\theta = \cos(\theta)$. The UAV quadrotor attitude equation is

$$\begin{cases} \ddot{\phi} = \dot{\theta}\dot{\psi}((I_{yy} - I_{zz})/I_{xx}) + (L(F_{L4} - F_{L2})/I_{xx}) \\ \ddot{\theta} = \dot{\phi}\dot{\psi}((I_{zz} - I_{xx})/I_{yy}) + (L(F_{L3} - F_{L1})/I_{yy}) \\ \ddot{\psi} = \dot{\phi}\dot{\theta}((I_{xx} - I_{yy})/I_{zz}) + (\sum_{j=1}^4 (-1)^j \gamma \omega_j^2 / I_{xx}) \end{cases} \quad (5)$$

where γ is the rotational air resistance coefficient, L denotes the distance from each rotor to the center of gravity of the quadcopter UAV, $F_{L1}, F_{L2}, F_{L3}, F_{L4}$ are the forces generated by the four motors of Motor1----Motor 4, I denotes the rotational moment of inertia of the quadcopter UAV in the fuselage coordinate system, and $\dot{\phi}, \dot{\theta}, \dot{\psi}$ and C are the angular velocities of the traverse roll, pitch, and yaw in the fuselage coordinate system. The complete nonlinear equation of the UAV is:

$$\begin{cases} \ddot{x} = (u_1/m)(s\psi s\varphi + c\psi c\varphi s\theta) - (k_1/m)\dot{x} + d_1 \\ \ddot{y} = (u_1/m)(s\psi s\theta c\varphi + c\psi s\varphi) - (k_2/m)\dot{y} + d_2 \\ \ddot{z} = (u_1/m)c\theta c\varphi - g - (k_3/m)\dot{z} + d_3 \end{cases} \quad \begin{cases} \ddot{\phi} = \dot{\theta}\dot{\psi}((I_{yy} - I_{zz})/I_{xx}) + (u_2/I_{xx}) + d_4 \\ \ddot{\theta} = \dot{\phi}\dot{\psi}((I_{zz} - I_{xx})/I_{yy}) + (u_3/I_{yy}) + d_5 \\ \ddot{\psi} = \dot{\phi}\dot{\theta}((I_{xx} - I_{yy})/I_{zz}) + (u_4/I_{zz}) + d_6 \end{cases} \quad (6)$$

where u controls the vertical motion of the quadcopter UAV; u controls the roll motion of the quadcopter UAV; u controls the pitch motion of the quadcopter UAV; u controls the yaw motion of the quadcopter UAV. d ($i=1,2,\dots,6$) are the system perturbations and the quadrotor modeling is completed.

2.2 UAV Simulink model

The parameters of the modeled UAV in matlab's simulink are shown in Fig 2.

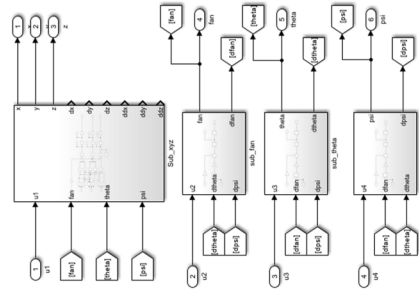


Fig. 2 Parameters of the model UAV

Some of these symbols are replaced with simple symbols as shown in the Table 1.

Table 1 symbols

Prime symbol	Substitute symbols in Simulink	Hidden meaning
ϕ	fan	Roll angle
θ	theta	Tilt
ψ	psi	Angle of divergence
ρ	rho	Quadcopter Lift Parameters
γ	gama	Air resistance parameters

3 Quadrotor UAV control system design

Quadcopter UAV control systems rely on PID controllers which are common feedback controllers that calculate the “proportional”, “integral”, and “differential” factors of the control

object to control the system. three factors to control the system. To do this, the controller is built in MATLAB/Simulink.

3.1 Position controller design and construction

According to the underlying flight control framework, the position controller accepts three position initial inputs, i.e. P_{xd} , P_{yd} , P_{zd} , and finally three outputs, i.e. θ_d , ϕ_d and F_d . The position controller is divided into two channels, the horizontal position channel and the overhead position channel⁶, whose input-output relationships are shown in Fig. 3.

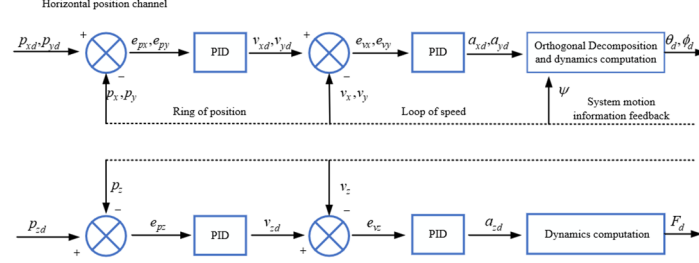


Fig. 3 Position controller input-output relationship design diagram

3.2 Attitude Controller Design and Construction

According to the underlying flight control framework, the attitude controller accepts three expected attitude inputs, i.e., θ_d , ϕ_d , and ψ_d , and the final three outputs, i.e., T_x , T_y , T_z . Two of the horizontal attitude angles correspond to the same channel parameters, and the yaw attitude angle parameter is set separately⁷. The input-output relationship is shown in Fig. 4.

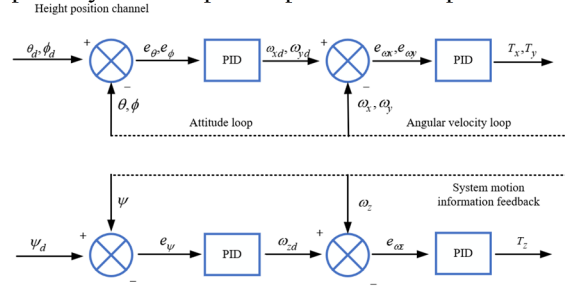


Fig. 4 Attitude controller input-output relationship design diagram

3.3 Control distributor design and construction

The control distributor interfaces the controller with the motors and is responsible for converting the previously obtained expected force and torque signals into four expected rotational speeds corresponding to the four propellers of the quadrotor in order to provide them to the subsequent motor controllers. The control distributor first accepts the expected lift force F_d provided by the altitude position channel in the position controller, as well as the three expected torques T_x , T_y , T_z provided by the attitude controller, and obtains the expected rotational speeds of the four propellers as subsystem outputs through matrix operations based on the propeller dynamics equations. Propeller dynamics equations:

$$\begin{aligned} F &= C_T(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \\ T_x &= (\sqrt{2}/2) d C_T(\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2) \\ T_y &= (\sqrt{2}/2) d C_T(\omega_1^2 - \omega_2^2 - \omega_3^2 + \omega_4^2) \\ T_z &= C_M(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{aligned} \quad (7)$$

Matrix form:

$$\begin{bmatrix} F \\ T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} c_T & c_T & c_T & c_T \\ (\sqrt{2}/2) d c_T & (\sqrt{2}/2) d c_T & -(\sqrt{2}/2) d c_T & -(\sqrt{2}/2) d c_T \\ (\sqrt{2}/2) d c_T & -(\sqrt{2}/2) d c_T & -(\sqrt{2}/2) d c_T & (\sqrt{2}/2) d c_T \\ c_M & -c_M & c_M & -c_M \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \quad (8)$$

$$\begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} = \begin{bmatrix} c_T & c_T & c_T & c_T \\ (\sqrt{2}/2)dc_T & (\sqrt{2}/2)dc_T & -(\sqrt{2}/2)dc_T & -(\sqrt{2}/2)dc_T \\ (\sqrt{2}/2)dc_T & -(\sqrt{2}/2)dc_T & -(\sqrt{2}/2)dc_T & (\sqrt{2}/2)dc_T \\ c_M & -c_M & c_M & -c_M \end{bmatrix} \begin{bmatrix} F \\ T_x \\ T_y \\ T_z \end{bmatrix} \quad (9)$$

The above arithmetic relations are ported to Simulink to complete the model building of the control distributor as shown in Fig. 5.

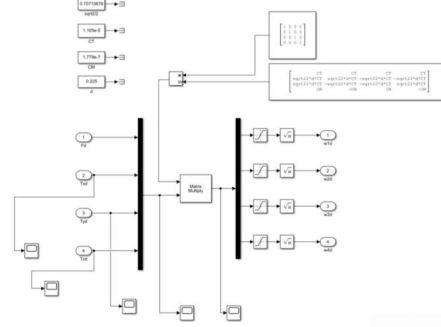


Fig. 5 Control distributor model

3.4 Design and construction of the motor controller

The motor controller accepts the desired speed from the control distributor as a system input and outputs the desired throttle command for each motor⁸. When controlling the motors, the relationship between motor speed and throttle command can be considered linear with the following relationship. $\omega_k = C_R \sigma + \omega_b$, where C_R and ω_b are the motor parameters, and so we

have $\sigma_d = \frac{1}{C_R}(\omega_{kd} - \omega_b)$. It is sufficient to use open-loop control, i.e., there is no need to introduce feedback. The model is built in Simulink as shown in Fig. 6

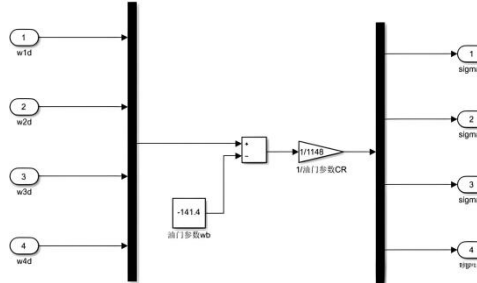


Fig. 6 Motor controller model

4 MATLAB-based UAV formation cooperative control

4.1 UAV formation control theory

Formation control theory is a key research area in distributed multi-intelligent body systems, which focuses on how to achieve cooperative movement and formation maintenance of multiple intelligences by designing effective control protocols.

Consistency theory, which is the basis of formation control, deals with how the interactions between intelligences can be used to ensure that their states achieve consistency. In this way, the whole group is able to maintain formation and perform synergistic tasks even when only local information is available⁹.

The virtual structure approach treats the formation as a whole in which the intelligences play specific roles to maintain the morphology of the whole formation by controlling their position in the structure. The leader-follower model is a common formation control strategy in which followers adjust their positions based on the leader's dynamics and the relative position information between them to maintain the formation.

Behavior-based methods mimic the group behavior of organisms in nature by providing the intelligences with a set of simple behavioral rules. These rules enable the intelligences to perform formation control autonomously without centralized control. Distributed sampling control and

event-triggered control are two advanced control methods that optimize the formation control process by reducing communication requirements.

4.2 Pilot-follower UAV formation cooperative control

Pilot-follower UAV formation cooperative control is a UAV cooperative flight technique. In this control mode, one or more pilot UAVs are responsible for guiding a group of follower UAVs so that they can maintain a predetermined spacing and relative position in the air, thus realizing formation flight¹⁰.

In the implementation of pilot-follower UAV formation cooperative control, suitable algorithms and control strategies need to be designed. sensors can be utilized to obtain information about the surrounding environment, and wireless data transmission and command interaction can be achieved with the help of communication systems. The code is created and simulation experiments are performed in MATLAB as shown in Figure 7.

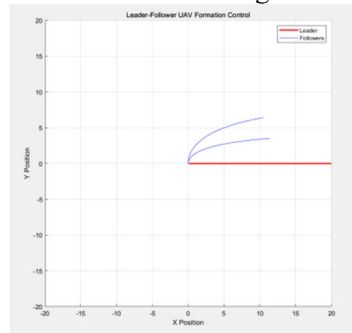


Fig 7 Pilot-Follower UAV formation cooperative control

5 Conclusion

In this paper, the control system design and formation cooperative control technology for quadrotor UAV formation are studied in depth, the quadrotor UAV control system based on PID controller is realized by MATLAB simulation platform, and the leader-follower formation cooperative control strategy is proposed. The stability of the designed control system and the effectiveness of the formation cooperative control strategy are verified through theoretical analysis and simulation experiments.

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